

Keynote · Multimodal Data Integration: From Biomarkers to Mechanisms

Presenter: Caroline Uhler — Broad Institute · MIT

Session: Day 2 · Fri Oct 2, 2026 · 9:00–9:45 AM · Morning keynote

An exciting opportunity at the intersection of the biomedical sciences and machine learning stems from the growing availability of large-scale multi-modal data (imaging-based and sequencing-based, observational and perturbational, at the single-cell level, tissue-level, and organism-level). Traditional representation learning methods, although often highly successful in predictive tasks, do not generally elucidate underlying causal mechanisms. I will present a statistical and computational framework for causal representation learning and its applications towards identifying novel disease biomarkers as well as inferring gene regulation in different disease contexts.